

Neurological Manifestations of Hantavirus Infection A Review

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IMPORTANCE Hantaviruses have caused several recent human disease outbreaks. They carry a high mortality rate, and some species can spread from person to person although they are typically spread from rodents to humans. Neurological complications have been documented but are poorly understood.

OBSERVATIONS Broadly, 2 types of syndromes have been associated with hantaviruses. The pulmonary syndrome carries a high mortality rate and can result in pulmonary and cardiac failure. Neurological involvement is typically secondary to hypoxic and ischemic injury or metabolic dysfunction. Another manifestation of the infection that is associated with some species of the virus is termed *hemorrhagic fever with renal syndrome*. Some patients may develop a meningoencephalitis or pituitary apoplexy resulting in sudden visual loss and pituitary dysfunction. Rarely, seizures, myelitis, and peripheral neuropathy may occur. Although there are no specific antiviral drugs approved for treatment of hantavirus, there are several that showed efficacy in preclinical trials. Management is supportive care and treatment of symptoms.

CONCLUSIONS AND RELEVANCE Neurological manifestations of hantavirus infection are uncommon but can be severe. Prospective studies and experimental models are needed to better characterize these manifestations, understand pathophysiology, identify therapeutic targets, and develop guidelines for management.

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The recent outbreak of hantavirus on a cruise ship that resulted in the death of 3 individuals and the infection of 8 others from a single infected person has brought back memories of SARS-CoV-2 infection, which also emerged on a cruise ship during the early days of the pandemic.¹ However, hantavirus does not carry the same risk of causing a widespread infection. A prompt response by the health authorities ensures that the infection will be contained. Because the virus results in periodic outbreaks and there may be a long latent period, neurologists need to be aware of the potential neurological manifestations of the infection for proper management of these individuals. The clinical manifestations may involve the central and peripheral nervous system. These patients may require acute and long-term neurological care. Besides, neurologists may play an important role in managing public fears and communication. They are also critical for pandemic preparedness and in conducting clinical studies.

Methods

A comprehensive English-language literature search was conducted using PubMed/MEDLINE and Embase databases and the World Health Organization and Centers for Disease Control and Prevention websites from inception to May 16, 2026. The search strategy combined MeSH terms related to hantavirus, orthohantavirus, Andes, Puumala, Hantaan, brain, meningitis, encephalitis, pituitary, spinal cord, myelitis, neuropathy, animal models, antivirals, treatment, immunotherapy, virus structure, epidemiology, ecology, vectors, reservoirs, pathogenesis, hantavirus pulmonary syndrome, vascular permeability, and hemorrhagic fever and renal syndrome.

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Epidemiology

Although the virus is most commonly present in rodents, it can also infect shrews, bats, and moles.² They are most commonly transmitted to humans through contact with infected urine, fecal droppings, and rodent saliva, often by inhaling contaminated dust. Rarely, infection can occur from rodent bites or scratches. The rodents develop a chronic infection with sustained viremia for prolonged periods with minimal effects.³ Approximately 10% to 20% of wild deer mice (the primary carrier in the US) carry hantavirus. Although infection in mice is relatively common in rural areas, human contraction of the disease is rare. From 1993, the year when national surveillance began, through 2023, a total of 890 laboratory-confirmed cases of hantavirus were reported in the US.⁴ Factors such as changes in climate and aspects of human behavior that aid the spread of the virus have been well studied.³ Those camping or living in rural areas or flood-prone regions; those clean-

ing sheds, barns, and cabins; and demolition and pest-control workers are at greater risk for exposure to rodents and thus to hantavirus.^{5,6}

Betsy Arakawa, the wife of actor Gene Hackman, contracted fatal hantavirus in early 2025 likely through exposure to infected rodent droppings or urine at their Santa Fe, New Mexico, home. Investigations revealed extensive rodent nesting in 8 outbuildings on their property.⁷

The most recent outbreak on the cruise ship in May 2026 was caused by the Andes virus. The virus was sequenced very quickly and has been identified as Andes virus clade 3.^{1,8} Andes virus is the only species of hantavirus shown to spread from person to person. An outbreak in Argentina in 2018 to 2019 resulted in 34 confirmed infections and 11 deaths.⁹ The virus is also endemic in the southern part of Chile. Mother-to-child transmission has also been documented via breast milk¹⁰ even though maternal antibodies are thought to protect against vertical transmission. This is also the species that infected the passengers recently on the cruise ship.¹

The R value of secondary human-to-human transmission is 2.⁹ In comparison, the R value for measles is 15 and for SARS-CoV-2 is 10. Seroprevalence studies suggest that it is likely that mild cases of hantavirus infection are underrecognized and underreported.¹¹ Even though human-to-human transmission is uncommon, close contact can lead to multiple horizontal transmissions among humans.⁹

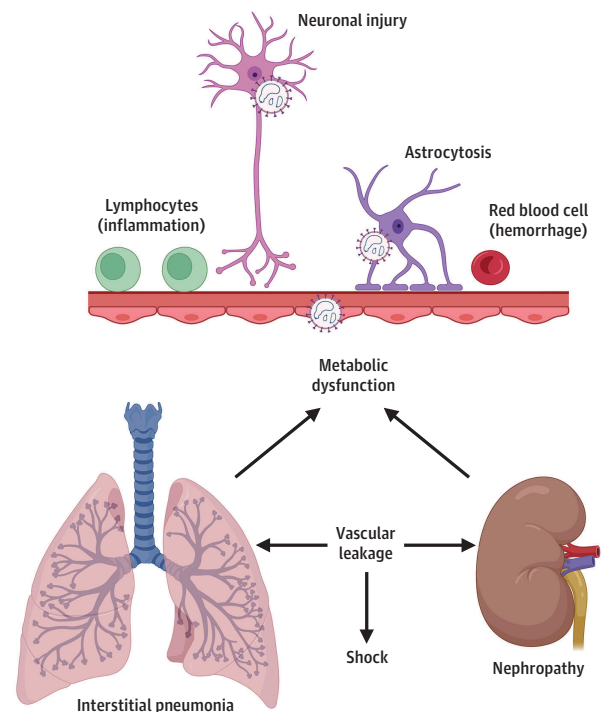
Hantavirus Structure

The virus belongs to the genus *Orthohantavirus*, family Hantaviridae. It is an enveloped, negative-sense, single-stranded RNA virus. It is 80 to 120 nm in diameter. The virus has a segmented genome with 3 RNA segments. The S (small) segment encodes the nucleocapsid (N) protein. The M (medium) segment encodes a glycoprotein precursor that is cleaved into glycoproteins Gn and Gc. The L (large) segment encodes the RNA-dependent RNA polymerase (RdRp). The key structural components of the virus are the lipid envelope that is derived from the host cell membrane and surrounds the virus particle. The surface glycoproteins, Gn and Gc, are embedded in the envelope. They form spikelike projections on the viral surface. These are important for attachment to host cells, entry or fusion, and immune recognition. The nucleocapsid is inside the envelope and coats each RNA segment. This forms helical ribonucleoprotein complexes. The RdRp is packaged with the viral RNA within the virus. It is needed for transcription and replication of the virus (Figure). The S segment also encodes a nonstructural protein that plays a role in evasion of the immune system.¹² The virus does not have any matrix protein. In vitro, the virus replicates slowly and is noncytopathic, making it difficult to develop high-throughput antiviral screening assays.¹³

Hantavirus Species

Since its discovery in the 1970s, the classification of Hantavirus and its taxonomy have continued to evolve. Currently 38 species of *Orthohantavirus* have been described, of which 24 can cause infection in humans and only 1 (Andes) has been shown to cause human-to-human transmission^{14,15} (Table 1). The geographic distribution of

Figure. Neuropathogenesis of Hantavirus Infection



The virus enters through the respiratory track, where it infects the epithelial cells. It spreads to infect the endothelial cells in most organs, leading to vascular leakage resulting in interstitial pneumonia, edema and may progress to respiratory failure. Infection of the kidney causes a nephropathy with proteinuria. The heart and lungs may be similarly involved. The ensuing metabolic dysfunction, hypoxia, and decreased cerebral perfusion can cause a metabolic encephalopathy. Animal studies suggest that the virus can infect brain endothelial cells, pericytes, astrocytes, and neurons. Vascular leakage can cause cerebral hemorrhage, perivascular inflammatory cells, neuronal apoptosis, and astrocyte activation resulting in a meningoencephalitis. Figure made using Biorender.

the hantavirus species parallels the distribution of the rodent species that it infects, except for the Seoul virus, which is worldwide because it infects *Rattus rattus* (the Black rat), a globally distributed organism³ (Table 1). Some species have been associated with more frequent neurological complications. Probably, Old World hantaviruses such as the Puumala virus, Dobrava-Belgrade virus, Hantaan virus, and Seoul virus are more often associated with central nervous system inflammatory syndromes like meningitis or meningoencephalitis, compared with New World hantaviruses.¹⁶⁻¹⁹ Of these viruses, Puumala virus has been most commonly reported in the literature to be associated with neurological manifestations (Table 1).²⁰ Infection with 1 hantavirus strain can provide broad protection against other strains of the virus²¹

Pathophysiology

In humans, the virus infects macrophages, dendritic cells, microvascular endothelial cells,²² and bronchial and bronchiolar epithelial cells, although there are large donor-specific variations.²³ The infection is noncytopathic; however, the endothelial barrier integrity

Table 1. Major Hantavirus Species and Their Hosts

Virus	Main reservoir host	Main region	Typical human syndrome
Old World viruses			
Hantaan virus	Striped field mouse (<i>Apodemus agrarius</i>)	China, Korea, Russia	HFRS, often severe
Seoul virus	Norway rat (<i>Rattus norvegicus</i>), black rat (<i>Rattus rattus</i>)	Worldwide (especially urban areas)	HFRS, usually mild to moderate
Puumala virus	Bank vole (<i>Myodes glareolus</i>)	Northern and Central Europe	Nephropathia epidemica (milder HFRS)
Dobrava-Belgrade virus	Yellow-necked mouse (<i>Apodemus flavicollis</i>), sometimes other <i>Apodemus</i> species	Balkans, Eastern/Central Europe	HFRS, can be severe
New World viruses			
Sin Nombre virus	Deer mouse (<i>Peromyscus maniculatus</i>)	North America	HPS
Andes virus	Long-tailed pygmy rice rat (<i>Oligoryzomys longicaudatus</i>)	Argentina, Chile	HPS; notable for possible person-to-person spread
Laguna Negra virus	Vesper mouse (<i>Calomys laucha</i>)	Paraguay, Bolivia, nearby regions	HPS
Black Creek Canal virus	Hispid cotton rat (<i>Sigmodon hispidus</i>)	Southeastern US	HPS
Bayou virus	Marsh rice rat (<i>Oryzomys palustris</i>)	Southern US	HPS
New York virus	White-footed mouse (<i>Peromyscus leucopus</i>)	Northeastern US	HPS
Puumala-like/other vole-borne hantaviruses	Various voles	Europe, Asia	Usually HFRS-like illness
Prospect Hill virus	Meadow vole (<i>Microtus pennsylvanicus</i>)	North America	Not a major recognized human pathogen

Abbreviations: HFRS, hemorrhagic fever with renal syndrome; HPS, hantavirus pulmonary syndrome.

is compromised due to excessive innate and adaptive immune responses leading to organ dysfunction.²⁴ Several mechanisms have been implicated in hantavirus-induced vascular permeability. This includes modulation of the expression of vascular endothelial growth factor, which enhances endothelial vascular permeability,²⁵ internalization, and degradation of vascular endothelial cadherin, an endothelial cell junction protein that is responsible for maintaining vascular barrier function,²⁶ activation of the plasma kallikrein-kinin system,²⁷ and infection of pericytes.²⁸ In the kidneys, this leads to proteinuria. The urinary excretion of interleukin 6 correlates with the amount of proteinuria suggesting the role of immune-mediated injury.²⁹ Hemophagocytic lymphohistocytosis associated with the Puumala virus infection has been described.³⁰

A wide variety of animal models for hantavirus infection have been developed, but none of them mimic human disease in part because the virus is mostly nonpathogenic or minimally pathogenic in rodents.¹² In rodents, chronic infection with hantaviruses is associated with regulatory T cell-mediated immunomodulation and subdued endothelial responses such that they rarely develop disease.³¹ In a macaque model of hantavirus infection, the animals developed hantavirus pulmonary syndrome, which was characterized by thrombocytopenia, leukocytosis, and rapid onset of respiratory distress caused by severe interstitial pneumonia. Viral RNA was detectable in all organs including the brain and spinal cord. In situ hybridization and immunohistochemistry showed that the virus infected endothelial cells in all organs, although the predominant pathology was in the lungs. Similar to humans, there was heterogeneity in clinical manifestations, and some macaques developed mild disease.³² In 1 study¹⁹ in which they infected mice with Hantaan virus, the mice developed predominantly an acute encephalitis manifested by paralysis and stimulus-sensitive seizures progressing to death. Histopathology showed vasculitis with infiltration of mononuclear cells,

neuronal apoptosis, and viral antigen in neurons.¹⁹ In another study³³ of younger animals, infection of astrocytes was observed. Differential cellular tropism of the various hantavirus species has also been reported. The Andes virus has broad tropism for all cell types. The Sin Nombre virus readily infects pulmonary endothelial cells, whereas the Hantaan virus robustly infects endothelial cells, cardiomyocytes, and astrocytes.³⁴

Clinical Manifestations

The incubation period can vary from 7 days to as long as 6 weeks.³⁵ It appears that the virus spreads only during the symptomatic phase of the infection. Initial symptoms are nonspecific, often leading to a delay in diagnosis. Two major clinical syndromes have been described with different neurological complications (Table 2).

Hantavirus Pulmonary Syndrome

Hantavirus pulmonary syndrome (HPS) is seen mainly in the Americas. It is usually of rapid onset with fever and muscle aches, then cough and shortness of breath. It can quickly progress to pulmonary edema, respiratory failure, and cardiogenic shock.^{3,36} Capillary leakage occurs in the lungs.³⁷ Early symptoms can include fever, fatigue, nausea, and muscle aches, especially in large muscle groups, which can be easily confused with influenza or other viral illnesses. Approximately half of all patients with HPS have experienced headaches, dizziness, chills, and gastrointestinal symptoms, including nausea, vomiting, diarrhea, and abdominal pain. Late symptoms appear 4 to 10 days after the initial phase of illness and include coughing, shortness of breath, and chest tightness. Individuals are generally only infectious while symptomatic. Early diagnosis of HPS can be difficult, especially within the first 72 hours of

symptoms, before the virus can be accurately detected in body secretions and excretions.

Neurologic complications are usually secondary to critical illness and related to hypoxia, cerebral hypoperfusion, and metabolic dysfunction; shock frequently results from severe capillary leakage. Bleeding from disseminated intravascular coagulation can rarely occur. In this situation, neurological symptoms may include headache, changes in mental status, delirium, and seizures. Symptoms related to cerebral hypoxia and global ischemia may also occur. Rarely, cerebral hemorrhages can occur. One case of postviral encephalitis has been described with the Andes virus. This patient developed HPS and encephalitis with altered mental status; seizures were noted as the patient was recovering from the pulmonary symptoms. He had an abnormal magnetic resonance imaging (MRI) scan, pleocytosis, and elevated protein concentration in the cerebrospinal fluid (CSF).³⁸ Other case reports of acute disseminated encephalomyelitis and transverse myelitis after HPS have also been described.³⁹

Hemorrhagic Fever With Renal Syndrome

The most common serotypes causing hemorrhagic fever with renal syndrome (HFRS) are Hantaan, Dobrava, Seoul, and Puumala. This is endemic in Eastern Russia and parts of Asia. HFRS manifests with fever and is associated with thrombocytopenia and an acute nephropathy.

Classically, HFRS occurs in 5 distinct phases: febrile, hypotensive, oliguric, polyuric, and convalescent. Pulmonary involvement and neurological manifestations may occur early in the course.³ Neurological complications are much more common with this illness.²⁰ In a review⁴⁰ of 811 individuals with HFRS due to Puumala virus, 97% had headaches, 40% had blurred vision, 31% had vomiting, 27% had all 3 symptoms, and 1% developed severe neurological manifestations. The severe manifestations included severe headache, meningitis, and, rarely, encephalitis. The virus can be detected in the CSF.⁴¹ Intracranial hemorrhage may occur due to thrombocytopenia. This often manifests as an intraparenchymal bleed, although subarachnoid and subdural bleeds may also occur.^{42,43} A unique manifestation is pituitary hemorrhage or infarction resulting in panhypopituitarism especially with Puumala and other Old World hantaviruses.⁴⁴ Metabolic encephalopathy may occur due to kidney insufficiency. Visual symptoms are also unique to this illness and can present as blurred vision, myopia, or eye pain with ocular symptoms in 70% of patients.⁴⁵ Rarely, patients may develop a peripheral neuropathy or strokes. An important aspect of this illness is that pituitary damage can cause long-term hormonal problems that may not manifest until weeks to months after the acute illness.⁴⁶ This may result in fatigue, sexual dysfunction, amenorrhea, hypothyroid symptoms, or secondary adrenal insufficiency.⁴⁷ Some may develop diabetes insipidus.^{48,49} In 1 study¹⁶ of 58 hospitalized patients with Puumala virus infection, most patients developed neurological manifestations, nearly half of them had CSF abnormalities, and 2 had sudden loss of vision due to pituitary apoplexy. Auto-immune hypophysitis with Puumala virus infection may also occur.⁵⁰ A case of focal encephalitis with Dobrava virus has been reported.¹⁷

Some patients can develop posterior reversible encephalopathy syndrome (PRES). It should be part of the differential diagnosis in patients who have severe hantavirus infection with kidney failure, hypertension, seizures, visual symptoms, or encephalopathy. MRI of the brain shows posterior vasogenic edema. Asymptomatic lesions involving the splenium of the corpus callosum have also been

Table 2. Neurological Manifestations of Hantavirus Syndromes

Feature	HPS	HFRS
Main organs affected	Lungs, cardiovascular system	Kidneys, vascular system
Common neurologic symptoms	Mild headache; severe neurological signs in critical illness	Headache common; broader range of neurologic complications
Encephalopathy	Due to hypoxia or shock	Due to uremia or bleeding
Seizures	Uncommon	Uncommon but better described
Intracranial hemorrhage	Rare	More likely due to thrombocytopenia and bleeding
Pituitary hemorrhage/hypopituitarism	Rare	Important complication
Visual symptoms	Rare	More common
Meningoencephalitis	Rare	More common

Abbreviations: HFRS, hemorrhagic fever with renal syndrome; HPS, hantavirus pulmonary syndrome.

described.⁵¹ A case of Seoul virus with meningitis has also been described.⁵²

Seizures occur secondary to electrolyte derangements, uremia, hypoxic injury, intracranial hemorrhage, PRES, or encephalitis.^{17,53} Acute disseminated encephalomyelitis with HFRS that typically follows the febrile illness can also occur.^{54,55} Some can develop an acute inflammatory demyelinating peripheral neuropathy that responds to treatment with intravenous immunoglobulin.⁵⁶⁻⁵⁸ A case of peripheral neuropathy with intractable neuropathic pain has also been described.⁵⁹ Longitudinally extensive transverse myelitis may also occur.³⁹ Hantaan virus has been isolated from the brain of an individual who died of HFRS.⁶⁰

Long-term consequences of the infection on the nervous system have not been well studied. In 1 study,⁶¹ with 20-year follow-up of Puumala virus infection-related nephropathia epidemica, a mild form of HFRS, 78% of patients had persistent hypertension, and 8% had chronic kidney disease.

Prognosis of Hantavirus Infection

Hantavirus has a high case fatality rate, often ranging from 35% to 50% for HPS in the Americas. The disease causes severe respiratory failure, with death occurring in 35% to 40% of US cases. HFRS in Asia and Europe has a 1% to 15% fatality rate. Prognostic factors include age, with higher mortality in adolescents.⁶ Viral load rather than duration of viremia is also an important factor.⁶² A meta-analysis⁶³ identified several risk factors associated with mortality. These included age over 18 years, female sex, rural residence, elevated creatinine levels, increased hematocrit concentration, signs of bleeding, and the presence of infiltrates on chest radiographs.⁶³

Management of Hantavirus Infection and Associated Complications

Diagnosis can be established by detection of virus by reverse transcriptase polymerase chain reaction or by detection of immunoglobulin M (IgM) and IgG antibodies to the virus. There is no spe-

cific Food and Drug Administration–approved antiviral drug against hantavirus, and there is no licensed vaccine available, although there are several vaccines in preclinical trials.^{64,65} There are also several potential broad antiviral drugs that could be explored in clinical trials. This may include drugs that inhibit RNA-dependent RNA polymerase inhibitors such as favipiravir and molnupiravir.⁶⁶ Other potential candidates are ribavirin, which is a nucleoside analog⁶⁷ and griffithsin, which is an entry inhibitor.⁶⁸ In vitro data are available for all of these compounds suggesting efficacy against hantavirus.⁶⁹ Infection with the virus leads to broadly neutralizing antibodies; therefore, clinical trials with convalescent plasma or intravenous immunoglobulin from regions where the virus is endemic should be considered. A nonrandomized multicenter clinical trial⁷⁰ of convalescent human plasma showed a decrease in case fatality rate in HPS. Neutralizing monoclonal antibodies to hantavirus are being developed.^{69,71} A double-blind placebo-controlled trial⁷² with high-dose methylprednisolone did not show any benefit in severe HPS. Preventive measures should include rodent control and minimizing exposure to rodent saliva and excreta.

In patients with vascular leakage, a bradykinin receptor antagonist, icatibant⁷³ or vandenatib, which targets vascular endothelial cell growth factor signaling, could be considered.⁷⁴

Because most neurological complications occur due to systemic manifestations, the key is to avoid anchoring on hantavirus alone. Immediate priorities should be to secure airway, oxygenation, and hemodynamics. Advanced critical care with mechanical ventilation and extracorporeal membrane oxygenation may be necessary. Blood glucose level, electrolyte levels, kidney function, and blood gases should be checked if unstable, in addition to complete blood count with platelets and a coagulation panel including fibrinogen if there is concern for bleeding. Liver function tests, creatinine kinase concentration, and lactate levels should be monitored. Due to the possibility of human-to-human transmission with the Andes virus, individuals who may have been exposed to the virus and are asymptomatic may self-quarantine for 6 weeks, and those who may be symptomatic or considered high-risk should be quarantined in high-level isolation units.⁷⁵

If altered mental status, seizure, focal deficit, thunderclap headache, or visual syndrome are present, a noncontrast computed tomography scan of the head should be obtained. If the patient is stable, a brain MRI should be obtained. MRI imaging of the pituitary gland should be considered if there is severe headache, visual change, hyponatremia, or hypotension suggesting apoplexy or pituitary hemorrhage. Due to pituitary necrosis, an MRI scan may show empty sella syndrome.¹⁸ Lumbar puncture may be performed if meningitis or encephalitis suspected, but only after brain imaging, if indicated, and after accounting for thrombocytopenia and coagulopathy. If pituitary involvement is suspected, the following should

be measured in the serum: morning cortisol level, adrenocorticotrophic hormone level, thyroid-stimulating hormone/free-T4 level, prolactin level, luteinizing hormone/follicle-stimulating hormone level, testosterone/estradiol level, and sodium/osmolality. In addition, empiric stress-dose steroids should be considered if adrenal crisis is a concern.

Gaps in Knowledge and Future Directions

Most of the knowledge about the neurological complications is derived from case reports, small series, and regional cohorts. The literature is biased toward description of severe and unusual manifestations. The infections often occur in rural settings where resources are limited; hence, it is likely that the cases are underreported in the literature. Natural history studies with prospectively collected data and long-term follow-up are necessary to fully understand the breadth and depth of involvement of the nervous system by the virus. Neurological or systemic manifestations in immunocompromised individuals are not known. It would be important to know if under these circumstances chronic or persistent infections could be established and if relapses may occur. Experimental studies are needed to understand the neurotropic properties of the virus and to study the neuropathogenesis of the infection. It also remains to be seen if the virus can adapt to humans with more efficient person-to-person transmission and if the virus can be shed in the asymptomatic phase of the infection. Animal studies, however, show that the Andes virus is genetically stable after serial passages.⁷⁶ Although it is important to develop antiviral agents, it is also important to determine at what stage of the infection anti-inflammatory agents might be beneficial.

Conclusions

As physicians, we also play a crucial role in communicating accurate information about the infection, addressing public fears and misinformation, and building trust with the patients, their families, and the public in general. We need to prevent the stigma directed toward patients and contacts while simultaneously communicating the importance of isolation in controlling the spread of infection.

It is equally important for us to develop prepandemic preparedness strategies. These may include the development of collaborative and trusted partnerships that have scientific agreements in place, meet regularly, and can be launched for the collection of information, surveys, biospecimens, and the development of a scalable framework for data collection and guidelines for management. Generic protocols for collection of data and research specimens from patients and contacts should be in place in preparation for potential pandemics.

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